

Richard A. Matos Arderí

COMPUTER SCIENTIST · DATA SCIENCE, ML & AI · FULL-STACK DEVELOPER

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LANGUAGES

Spanish	Native
English	C1 · Advanced
French	A2 · Elementary

HONORS & AWARDS

- ◆ **Scientific Merit Prize**
Cuban Higher Education · 2026
- ◆ **1st Place · Honors Examination in Ordinary Differential Equations**
MATCOM, UH · 2026
- ◆ **“Relevant” paper at BIOQUIM 2025**
Hybrid modeling of leptospirosis with vaccination

RESEARCH INTERESTS

- Biomedical AI
- Epidemiological Modeling
- Vaccine Development
- Mathematical Biology
- Distributed Systems

PROFILE

Computer Scientist trained at MATCOM (University of Havana), specializing in Data Science, Machine Learning and Artificial Intelligence, with strong full-stack and web development experience. Awarded the Scientific Merit Prize for an outstanding research record. Currently a Consultant Specialist at the Finlay Vaccine Institute (BioCubaFarma), where I design AI-powered platforms — multi-agent systems, RAG pipelines and AutoML workflows — and ship web, mobile and desktop applications. I take products end to end, from data and mathematical modeling through to polished, deployable software across web, Android, iOS and desktop.

TECHNICAL SKILLS

Programming Languages

Python · Rust · TypeScript / JavaScript · C · C++ · C# · R · Haskell · Prolog · SQL · LaTeX

Data Science & ML

scikit-learn · PyTorch · TensorFlow · XGBoost · LightGBM · CatBoost · SHAP · LIME · MLflow · NumPy · Pandas · SciPy · SymPy

AI / Agents / NLP

LangChain · LangGraph · RAG · Multi-agent systems · AutoML (FLAML) · n8n · HuggingFace · Ollama · ChromaDB · Ontologies & SPARQL (rdflib)

Modeling & Simulation

ODE / SEIR models · Agent-based modeling (Mesa) · Cellular automata · Markov chains (CTMC) · Metaheuristics (GA, SA, Tabu, GRASP) · Sensitivity analysis (SALib/Sobol)

Databases

PostgreSQL · MySQL · SQL Server · MongoDB · Redis · SQLAlchemy / Alembic

Web & Full-Stack

React · Next.js · Node.js · FastAPI · Django · Tailwind CSS · REST APIs · WebSockets

Mobile & Desktop

Flutter · Android · iOS · React Native · Electron / Tauri

Big Data & Distributed

Hadoop · HDFS · PySpark · Celery · Redis · Distributed crawling & indexing

DevOps & Tools

Docker · Docker Compose · Git · GitHub Actions (CI/CD) · CodeQL · Linux

QA & Testing

pytest · unittest · Selenium · Cypress · Playwright · Postman · JMeter · TDD · SonarQube · ruff / black / mypy

PROFESSIONAL EXPERIENCE

Consultant Specialist

Mar 2025 – Present

Finlay Vaccine Institute (IFV) · BioCubaFarma

Data-science and AI consulting for vaccine development: epidemiological modeling, clinical-data integration and auditing, and the design of predictive models and supporting software within IFV and BioCubaFarma projects.

EDUCATION

B.Sc. in Computer Science

Jan 2023 – Jun 2026

Faculty of Mathematics and Computer Science (MATCOM), University of Havana

GPA: 4.71 / 5.00 · Cohort C411.

Diploma in Clinical Trials

2025

Center for Genetic Engineering and Biotechnology (CIGB) · BioCubaFarma

Specialized postgraduate training in the design, conduct, regulation and data management of clinical trials.

International Winter School

2026

University of Havana

Intensive program of international scientific training and exchange.

FEATURED PROJECTS

FinlAI

AI · AutoML · Health

AI platform for optimizing vaccine development through integrated data management at the Finlay Vaccine Institute: multi-agent systems, RAG pipelines and AutoML workflows over clinical-trial data.

Hybrid Modeling

ODE · ML · Epidemiology

System for epidemiological modeling combining differential-equation models with machine learning; applied to leptospirosis transmission and the evaluation of vaccination strategies.

AuditSoft

Audit · Clinical data

Software for auditing clinical-trial data, ensuring quality, traceability and integrity of information within the IFV regulatory framework. Presented at COMPUMAT 2025.

ML Studio

AutoML · Predictive models

Machine-learning modeling environment for predicting in-hospital mortality from AMI and performing inverse optimization of clinical variables, with interpretability (SHAP) and validation.

Simmunity

ABM · ODE · FastAPI · Distributed

Computational simulator of the human immune system: agent-based, ODE and stochastic (Gillespie) models, a FastAPI REST layer, and distributed computing with Celery/Redis.

Clinical Trial DB Design

PostgreSQL · ER Modeling

Relational database design (13+ tables, 3NF) for the ISCANI clinical trial of the VCN7-T vaccine: patients, randomization, adverse events and regulatory audit trail.

climaXtreme

PySpark · HDFS · Docker

Analysis and modeling of extreme climate events over 8.6M+ records with Hadoop and PySpark; Streamlit dashboard, ML predictions and full Docker deployment.

HULK-Compiler-RS

Rust · Compilers

A complete compiler for the HULK language in Rust: lexical, syntactic and semantic analysis, and garbage collection, focused on performance and safety.

DistriSearch

Python · Distributed

Distributed search engine with distributed crawling, indexing and query processing (Distributed Systems final project).

CONFERENCE CONTRIBUTIONS

- 1** “Hybrid Modeling: A system for epidemiological modeling”
XLI Scientific Conference, Faculty of Physics, UH · May 2026
- 2** “AuditSoft: Software for auditing clinical-trial data”
COMPUMAT 2025 · International
- 3** “ODE Mathematical Model for leptospirosis transmission”
COMPUMAT 2025 · International
- 4** “Hybrid Modeling of leptospirosis transmission with vaccination strategies”
BIOQUIM 2025 · National · “Relevant” distinction
- 5** “Interactive Web Application for predicting Acute Myocardial Infarction”
BIOQUIM 2025 · National · with A. Ponce González
- 6** “ML models for in-hospital AMI mortality prediction and inverse optimization of clinical variables”
BIOQUIM 2025 · National · with A. Ponce González, A. Romero Imbert

RESEARCH PROJECTS

Optimizing vaccine development through integrated data management and AI

Finlay Vaccine Institute (IFV) · BioCubaFarma

Data Ecosystem Evolution Project

DICEI · IFV · BioCubaFarma

Development of a novel vaccine against Leptospirosis

IFV · BioCubaFarma

AMI registry with variable homogenization and quality indicators in Cuba

“Dr. Ernesto Guevara de la Serna” Teaching Hospital · MINSAP

VLIR-UOS: Integrated Real-Time Dengue Surveillance in Cuba for Outbreak Prediction

VLIR-UOS international cooperation